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OPTICAL MUSIC RECOGNITION FOR JAZZ LEAD SHEETS: A CRNN-CTC APPROACH FOR MONOPHONIC LINES

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GEP Deliverable 2: Temporal Planning

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1 Project Overview

This document presents the temporal planning for the project *Optical Music Recognition for Jazz Lead Sheets: A CRNN-CTC Approach for Monophonic Lines*. The planning follows the agile, iterative methodology described in Deliverable 1, organised into five sprints of ~ 4 weeks each plus a concurrent project management work package (WP0). Sprints 3 and 4 form a **refinement loop**: after initial training and evaluation (Sprint 3), error analysis feeds back into data pipeline and model improvements (Sprint 4).

The project runs from February to June 2026 (21 weeks), with a total estimated effort of **509 hours** at 4–5 h/day. It is carried out by a single developer who fulfils all roles, with biweekly advisory meetings (~ 1.5 h each) with the thesis supervisor. **Common resources** for all tasks: personal computer; Python 3.14, PyTorch, OpenCV, music21, LilyPond, L^AT_EX, Git/GitHub. Task-specific resources are noted below.

2 Task Description

Each task is identified by a code, assigned an estimated effort in hours, and linked to its dependencies and any specific resources beyond the standard set.

2.1 WP0: Project Management (130 h – concurrent)

T0.1 - Context and scope definition (20 h)

Research and write the project context, state of the art, scope, requirements, risks, and methodology (GEP Deliverable 1). *Dep*: none. *Res*: Zotero.

T0.2 - Temporal planning (12 h)

Work breakdown structure, effort estimates, dependencies, and Gantt chart (this document). *Dep*: T0.1.

T0.3 - Budget and sustainability report (15 h)

Economic budget and sustainability analysis (GEP Deliverable 3). *Dep*: T0.2.

T0.4 - Final GEP document integration (8 h)

Compile and revise GEP deliverables 1–3 into the final management document. *Dep*: T0.3.

T0.5 - Supervisor meetings (25 h)

Biweekly meetings (~ 1.5 h each incl. preparation), ~ 15 sessions. Concurrent with all tasks.

T0.6 - Thesis writing (50 h)

Continuous writing of the thesis document; intensifies during Sprint 5. Concurrent with all tasks.

2.2 WP1 – Sprint 1: Literature Review & Baseline (78 h)

Weeks 1–4 (February–March). Goal: establish the theoretical foundation and implement a classical CV baseline.

T1.1 - OMR literature survey (25 h)

Review traditional pipelines [1], CRNN-CTC [2], and transformer approaches [3], [4]. *Dep*: none. *Res*: IEEE/ACM/arXiv access.

T1.2 - Dataset exploration (15 h)

Inspect public OMR datasets (PrIMuS, CameraPrimus, MUSCIMA++, DeepScoresV2, OLiMPiC, GrandStaff); assess suitability. *Dep*: T1.1 (partial overlap[†]).

T1.3 - Classical CV baseline (25 h)

Morphological staff-line detection/removal with OpenCV; evaluate on clean, distorted, and scanned images. *Dep*: T1.2.

T1.4 - Baseline evaluation (10 h)

Quantitative evaluation (recall, precision, artefact rate) and failure-mode analysis. *Dep:* T1.3.

T1.5 - Sprint 1 review (3 h)

Present results to supervisor; confirm Sprint 2 scope. *Dep:* T1.4.

2.3 WP2 – Sprint 2: Dataset Construction (73 h)

Weeks 5–8 (March). Goal: build a synthetic monophonic dataset in Real Book style for CRNN-CTC training.

T2.1 - Output encoding design (12 h)

Define LMX-based vocabulary (pitch, octave, duration, rests, accidentals, barlines, clef/key/time signatures, CTC blank). Write PrIMuS-to-LMX conversion. *Dep:* T1.5. *Res:* music21.

T2.2 - Synthetic rendering pipeline (35 h)

LilyPond pipeline rendering ~ 43 k staff-line images (PNG) in jazz style (LilyJAZZ) with paired LMX annotations. *Dep:* T2.1. *Res:* LilyPond + LilyJAZZ.

T2.3 - Data augmentation (15 h)

Gaussian noise/blur, perspective warp to simulate scanned degradation, calibrated against real Real Book scans. *Dep:* T2.2. *Res:* OpenCV.

T2.4 - Validation and splits (8 h)

Validate image-label alignment, filter degenerate samples, prepare 80/10/10 splits. *Dep:* T2.3.

T2.5 - Sprint 2 review (3 h)

Review dataset quality with supervisor. *Dep:* T2.4.

2.4 WP3 – Sprint 3: CRNN-CTC Model Development (75 h)

Weeks 9–12 (April). Goal: design, implement, and perform the initial training run; establish the evaluation framework.

T3.1 - CRNN architecture design (25 h)

CNN backbone + bidirectional LSTM + CTC head in PyTorch; lightweight for consumer GPU. *Dep:* T2.5. *Res:* consumer GPU.

T3.2 - Training pipeline (20 h)

Training loop (CTC loss, LR scheduler, gradient clipping, early stopping, checkpointing) and data loader with dynamic padding. *Dep:* T3.1.

T3.3 - Initial training run (15 h)

First full training run; monitor convergence; document baseline metrics. *Dep:* T3.2. *Res:* GPU.

T3.4 - Evaluation framework (12 h)

Greedy CTC decoding, SER computation, per-sample error breakdown, worst-prediction visualisation. Partial overlap with T3.3. *Dep:* T3.2[†].

T3.5 - Sprint 3 review (3 h)

Analyse results with supervisor; plan Sprint 4 refinement. *Dep:* T3.3, T3.4.

2.5 WP4 – Sprint 4: Iterative Refinement (68 h)

Weeks 13–16 (April–May). Goal: close the train-evaluate-improve loop until SER stabilises. This sprint implements the iterative refinement cycle: after each run, errors are analysed, root causes identified, and targeted fixes applied before retraining.

T4.1 - Error analysis (10 h)

Breakdown by error type and token; identify systematic patterns. *Dep:* T3.5.

T4.2 - Data pipeline improvements (15 h)

Fix annotation errors, filtering, vocabulary adjustments; re-generate data as needed. *Dep:* T4.1.

T4.3 - Model refinement runs (25 h)

Hyperparameter adjustment and iterative retraining until SER stabilises. *Dep:* T4.2. *Res:* GPU.

T4.4 - Ablation studies (15 h)

Controlled experiments isolating key design choices; structured ablation table. *Dep:* T4.3. *Res:* GPU.

T4.5 - Sprint 4 review (3 h)

Present metrics and ablation results; decide on further iteration vs. final evaluation. *Dep:* T4.4.

2.6 WP5 – Sprint 5: Final Evaluation & Defence (85 h)

Weeks 17–21 (May–June). Goal: definitive evaluation, conditional extension, thesis completion, and defence preparation.

T5.1 - Final evaluation (15 h)

Evaluate best model on held-out test set; per-category SER; compare against CV baseline; publication-quality figures. *Dep:* T4.5. *Res:* GPU.

T5.2 - Qualitative error analysis (10 h)

Case studies of correct/incorrect predictions; failure-mode analysis by difficulty. *Dep:* T5.1.

T5.3 - Extension exploration – conditional (20 h)

If SER < 10% and time permits: prototype chord recognition or page-level staff detection. **Scope-contingent**; may be reduced or skipped. *Dep:* T5.1. *Res:* GPU.

T5.4 - Final thesis writing (30 h)

Complete all chapters, integrate figures/tables/references, proofread. *Dep:* T5.2 (and T5.3 if completed). *Res:* L^AT_EX, Zotero.

T5.5 - Defence preparation (10 h)

Slides, rehearsal, anticipated questions. *Dep:* T5.4.

2.7 Effort Summary

Table 1 summarises all tasks. Total: **509 h**. [†]Partial overlap (soft dependency).

3 Gantt Chart

Figure 1 shows the project schedule. WP0 runs concurrently throughout; each sprint ends with a review milestone (red diamond). Sprints 3 and 4 form the train–evaluate–improve cycle.

Table 1: Task summary with estimated effort, dependencies, and resources.

Code	Task	h	Dependencies	Resources
WP0 – Project Management (130 h)				
T0.1	Context and scope definition	20	-	Zotero
T0.2	Temporal planning	12	T0.1	-
T0.3	Budget and sustainability	15	T0.2	-
T0.4	Final GEP integration	8	T0.3	-
T0.5	Supervisor meetings	25	Concurrent	-
T0.6	Thesis writing	50	Concurrent	L ^A T _E X
WP1 – Sprint 1: Literature Review & Baseline (78 h)				
T1.1	OMR literature survey	25	-	Library access
T1.2	Dataset exploration	15	T1.1 [†]	-
T1.3	Classical CV baseline impl.	25	T1.2	OpenCV
T1.4	Baseline evaluation	10	T1.3	-
T1.5	Sprint 1 review	3	T1.4	-
WP2 – Sprint 2: Dataset Construction (73 h)				
T2.1	Output encoding design	12	T1.5	music21
T2.2	Synthetic rendering pipeline	35	T2.1	LilyPond
T2.3	Data augmentation	15	T2.2	OpenCV
T2.4	Validation and splits	8	T2.3	-
T2.5	Sprint 2 review	3	T2.4	-
WP3 – Sprint 3: CRNN-CTC Development (75 h)				
T3.1	CRNN architecture design	25	T2.5	Consumer GPU
T3.2	Training pipeline impl.	20	T3.1	-
T3.3	Initial training run	15	T3.2	GPU
T3.4	Evaluation framework	12	T3.2 [†]	-
T3.5	Sprint 3 review	3	T3.3, T3.4	-
WP4 – Sprint 4: Iterative Refinement (68 h)				
T4.1	Error analysis	10	T3.5	-
T4.2	Data pipeline improvements	15	T4.1	-
T4.3	Model refinement runs	25	T4.2	GPU
T4.4	Ablation studies	15	T4.3	GPU
T4.5	Sprint 4 review	3	T4.4	-
WP5 – Sprint 5: Final Evaluation & Defence (85 h)				
T5.1	Final evaluation	15	T4.5	GPU
T5.2	Qualitative error analysis	10	T5.1	-
T5.3	Extension exploration (cond.)	20	T5.1	GPU
T5.4	Final thesis writing	30	T5.2	L ^A T _E X
T5.5	Defence preparation	10	T5.4	-
TOTAL		509		

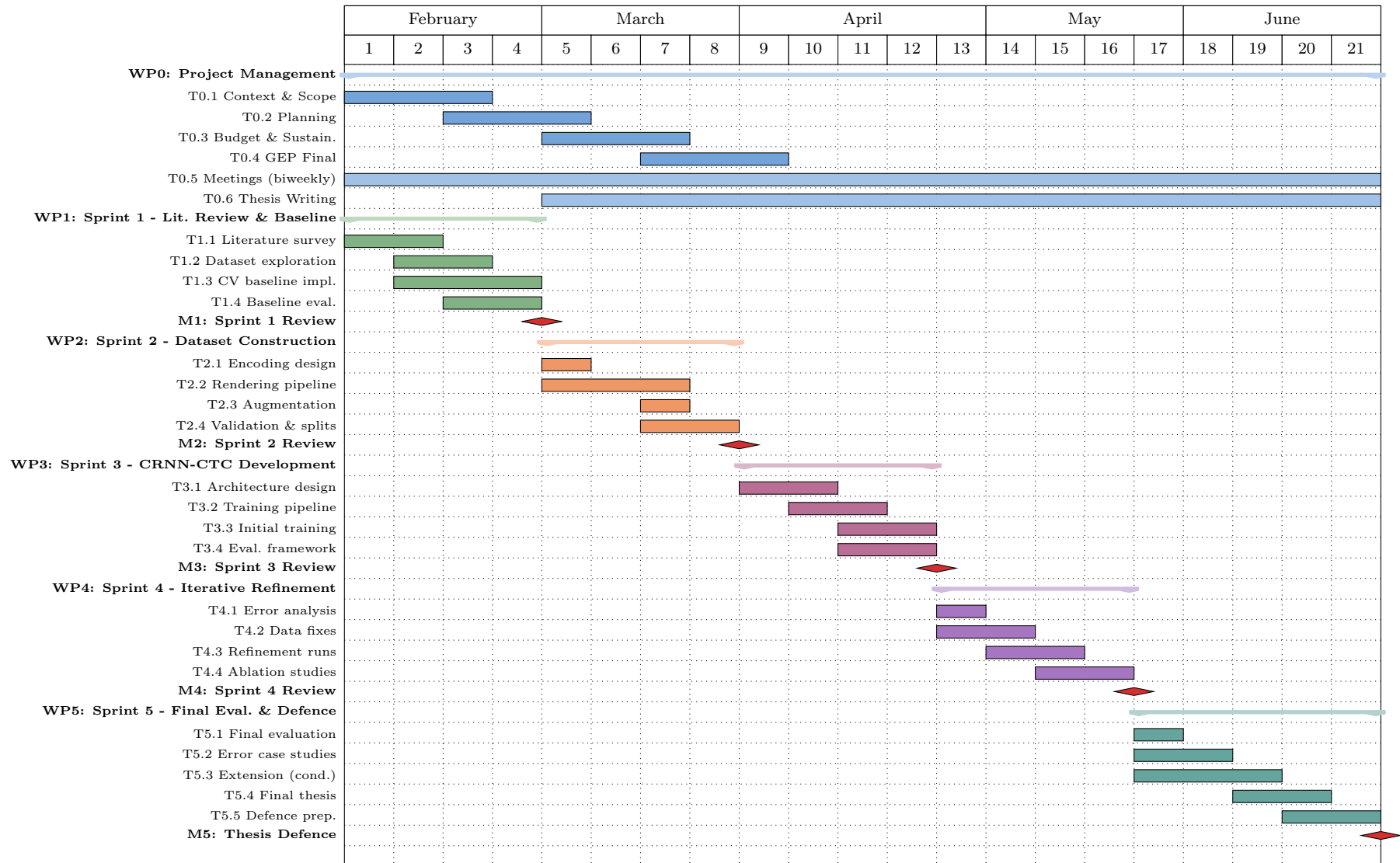


Figure 1: Project Gantt chart (weeks 1–21, February–June 2026). Coloured bars indicate work packages; red diamonds mark sprint reviews and the final defence milestone. WP0 runs concurrently throughout.

4 Risk Management and Contingency Plans

Table 2 summarises the five main risks, their contingency actions, schedule impact, and resource implications. These risks coordinate with those identified in Deliverable 1 (Section 3.6).

Table 2: Risk register with contingency plans. L=Likelihood, I=Impact (H=High, M=Medium, L=Low).

Risk	L/I	Contingency (Plan B)	Schedule impact	Add. resources
R1: Limited ground-truth data	H/H	Fall back to PrIMuS typeset images without jazz rendering (skip T2.2).	Saves ~35 h; ~10 h extra augmentation.	None.
R2: Staff detection errors	M/M	Restrict to manually cropped regions; affects only T5.3.	T5.3 hours → T5.4.	None.
R3: Model underfitting / CTC misalignment	M/H	Simplify: pitch-only vocab, clean images, or larger image height (within T4.2–T4.3).	Up to 10 h from T5.3 to refinement.	Longer GPU time.
R4: Computational constraints	L-M/M	Reduce data, simplify model, or use cloud GPU.	Up to 5 h cloud setup.	Cloud GPU (~€20–50).
R5: Scope creep	M/H	Reduce T5.3 to feasibility discussion (2–3 h writing).	Up to 18 h freed to T5.4.	None.

Built-in schedule buffers: overestimation in T2.2 and T4.3 (+5 h each); conditional extension T5.3 (20 h) as flexible buffer; concurrent thesis writing (T0.6). Total redistributable slack: ~25–30 h.

References

- [1] C. Dalitz, M. Droettboom, B. Pranzas, and I. Fujinaga, “A Comparative Study of Staff Removal Algorithms,” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 30, no. 5, pp. 753–766, May 2008, ISSN: 0162-8828. DOI: [10.1109/TPAMI.2007.70749](https://doi.org/10.1109/TPAMI.2007.70749). Accessed: Feb. 18, 2026. [Online]. Available: <http://ieeexplore.ieee.org/document/4359356/>.
- [2] “PrIMuS dataset,” Accessed: Feb. 18, 2026. [Online]. Available: <https://grfia.dlsi.ua.es/primus/>.
- [3] A. R. Vila, *Antoniiorv6/SMT*, Feb. 14, 2026. Accessed: Feb. 19, 2026. [Online]. Available: <https://github.com/antoniiorv6/SMT>.
- [4] “Ufal/olimpic-icdar24: Practical End-to-End Optical Music Recognition for Pianoform Music,” Accessed: Feb. 18, 2026. [Online]. Available: <https://github.com/ufal/olimpic-icdar24?tab=readme-ov-file>.